

Fairness-Transparency Trade-Off: How does increasing fairness in an interpretable model influence accuracy?

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1 Introduction

Artificial Intelligence (AI) has become increasingly prominent across many domains, including healthcare and security. This essay focuses specifically on healthcare, where Explainable AI (XAI) is of particular importance given the highly sensitive nature of the data worked with. AI models are increasingly being used to support diagnostic and treatment decisions; lack of transparency in how these models arrive at their conclusion can have serious consequences for patient outcomes and trust in the system. Thus, an AI system must not only perform well, but also be interpretable and fair across different patient groups.

A key concern in this context is whether striving for interpretability comes at the cost of predictive accuracy or fairness, therefore the research question discussed throughout this report is: how does the choice of fairness metric affect the perceived fairness and interpretability of an AI model in a healthcare setting.

Fairness is a concept especially critical in healthcare, as biased predictions can lead to unequal treatment across demographic groups, such as differences based on sex, age, and ethnicity (Carey et al., 2024). To investigate this, two different fairness metrics are examined: Equalized Odds and Positive Predictive Value (PPV) parity. Equalized Odds requires that a model produces equal True Positive Rates (TPR) and False Positive Rates (FPR) across groups, while PPV parity requires the reliability of positive predictions is consistent across groups. These two metrics were selected because they capture different dimensions of fairness, allowing for a more in-depth understanding of how differentiating fairness metrics shape the interpretation of model behavior.

2 Methods

2.1 Description of Fairness Metrics

PPV parity focuses on the reliability of positive predictions, and represents the probability that a positive prediction is correct. In other words, it captures how often the model is right when it predicts that an individual belongs to the positive class. Furthermore, PPV parity requires this probability to be equal across demographic groups, which means that if the model flags a patient as high risk in a clinical setting, the likelihood that this prediction is accurate should be consistent regardless of the patient's

demographic group. As positive predictions often trigger follow-up actions such as additional tests, imaging, or specialist referrals, ensuring equal PPV across groups is crucial. If PPV differs, some groups may receive less reliable alerts, leading to unnecessary procedures for some patients and missed opportunities for others (Nigam, 2025).

The other fairness metric discussed, Equalized Odds, approaches fairness by examining how a model distributes its errors across groups. It requires that both the TPR and the FPR are comparable for each group. This is due to the TPR reflecting how often the model correctly identifies individuals who truly have the condition, while the FPR captures how often it incorrectly flags individuals who do not. By focusing on these two error types, this fairness metric highlights whether certain groups face higher risks of missed detections or unnecessary follow-up (Mayer & Wilber, 2023). When looking at clinical prediction tasks, this distinction matters. Missing a true case can delay essential treatment, whereas a false alarm can trigger avoidable stress, additional testing, and potential harm. Thus, evaluating equalized odds reveals whether the model exposes groups to similar levels of benefit and risk.

2.2 Evaluation of Fairness Metrics

While both PPV parity and Equalized Odds aim to promote fairness, they focus on fundamentally different aspects of model behavior. PPV parity is concerned with the reliability of positive predictions, whereas Equalized Odds is concerned with the distribution of errors across groups. This distinction has important consequences in practice. For instance, it is mathematically impossible to simultaneously satisfy PPV parity, Equalized Odds, and demographic parity (Kleinberg et al., 2016). This is particularly relevant in healthcare, where disease prevalence frequently varies across demographic groups. When regarding patients with lung cancer, demographic groups are not the main priority, although they can have an effect. Differences in smoking history, environmental exposure, and access to screening are more likely to have an impact; therefore enforcing some fairness criteria come at the expense of others. These conflicts illustrate that fairness metrics cannot be applied mechanically, but instead require value-driven choices about which disparities matter most in a given clinical context.

Evaluating these metrics in practice involves computing group-specific performance measures and examining the size of the gaps between them. For PPV parity, one compares precision across groups, while for Equalized Odds, one compares the TPR and FPR (Nigam, 2025), (Mayer & Wilber, 2023). Visualizations such as grouped bar charts or group-specific ROC curves can help reveal where disparities arise and how they shift as classification thresholds change. Importantly, threshold adjustments designed to improve one metric often worsen another, meaning that improving PPV parity may reduce the TPR for some groups, while enforcing Equalized Odds may reduce PPV for groups with lower prevalence (Corbett-Davies et al., 2023). This confirms that fairness metrics are not merely diagnostic tools; they actively shape model design and influence clinical decisions.

2.3 Limitations of Fairness Metrics

Both metrics also have important limitations. Equalized Odds is difficult to implement in practice. Achieving it often requires complex mitigation strategies such as adversarial learning, reweighting, or post processing. These methods can reduce overall accuracy and may not generalize well to new data. Moreover, it also depends on the assumption that the ground truth labels are unbiased (Raftopoulos et al., 2025). In healthcare,

labels often reflect historical practices that may themselves be biased or incomplete. If the labels are biased, equalized odds may faithfully reproduce these biases rather than correct them. Lastly, the metric is also sensitive to dataset shifts, meaning that if the distribution of patients changes in deployment, the fairness guarantees may no longer hold (Emergent Mind, 2026).

PPV parity also has important drawbacks. Because of known impossibility results, it cannot be achieved at the same time as equalized odds when groups have different base rates (Saravanakumar, 2021). Moreover, it can create a misleading sense of fairness, as it focuses on calibration rather than on how errors are distributed. Thus, a model may satisfy PPV parity while still producing unequal false positive or false negative rates. As a result, though one group may be labeled as high risk, PPV parity may label them as equally accurate despite this not being the case (Fraenkel, 2020).

PPV parity can also hide poor performance within smaller subgroups. A model may look fair when evaluated on broad demographic categories but still perform badly for minority populations. These issues show that PPV parity should be interpreted carefully and used alongside other fairness metrics (Saravanakumar, 2021).

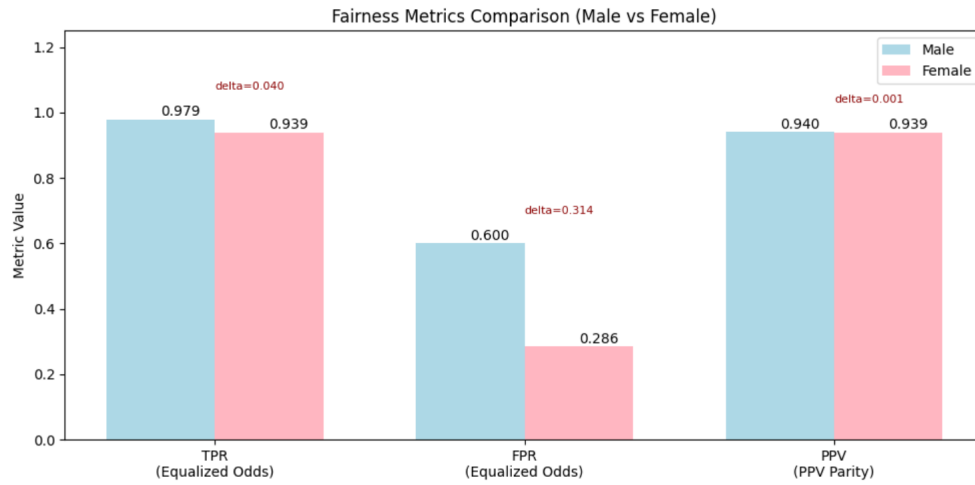
3 Code Demonstration

The dataset used for the code demonstration accompanying this analysis encompasses lung cancer diagnoses, with sex as the protected variable. Before starting the analysis, please find the link to the github page, containing a README, as well as the dataset and .ipynb file: https://gitlab.com/lea_rado_s1068713/xai_final_assignment/-/tree/cec095388083fc4260e0a8666629fa6584e13bbf/

The results reveal that the model behaves differently across the two fairness metrics examined. Beginning with Equalized Odds, the TPR disparity between males (0.979) and females (0.939) is small (difference = 0.040), falling just within the 0.05 fairness threshold, suggesting the model identifies true cancer cases at a broadly similar rate across both sexes. However, the FPR tells a notably different story, as males have an FPR of 0.600 compared to 0.286 for females, producing a large disparity of 0.3143, which clearly violates the fairness threshold. This means that healthy male patients are flagged as having lung cancer at a much higher rate than healthy female patients, indicating that the model is significantly more prone to false alarms for men.

In contrast, PPV parity is almost perfectly satisfied, with male PPV at 0.9400 and female PPV at 0.9394, resulting in a negligible difference of 0.0006. This indicates that when the model does predict cancer, that prediction is equally reliable regardless of the patient’s sex.

Taken together, these results illustrate a tension that is well documented in the fairness literature: a model can satisfy one fairness criterion while simultaneously violating another. Here, PPV parity is met, yet Equalized Odds is violated through the FPR. In a healthcare context, this distinction is clinically meaningful, as a high FPR for male patients could result in unnecessary follow-up procedures, increased patient anxiety, and a misallocation of medical resources.



4 Conclusion

In conclusion, fairness metrics provide valuable explanations of how predictive models behave across demographic groups. PPV parity and Equalized Odds highlight different aspects of fairness and reflect different ethical priorities. PPV parity focuses on the reliability of positive predictions, while equalized odds focuses on the distribution of errors. Their incompatibilities show that fairness is not a single objective but a set of competing values. In lung cancer risk prediction these metrics help reveal disparities that may otherwise remain hidden and support more informed decisions about model design and deployment. Fairness in healthcare requires careful consideration of context, values, and tradeoffs. Metrics can guide this process, but they cannot replace the need for thoughtful judgment.

5 References

- Carey, S., Pang, A., & de Kamps, M. (2024, September). Fairness in AI for healthcare. ScienceDirect. <https://www.sciencedirect.com/science/article/pii/S2514664524015674>
- Corbett-Davies, S., Gaebler, J. D., Nilforoshan, H., Shroff, R., & Goel, S. (2023, August 14). The measure and mismeasure of fairness. arXiv.org. <https://arxiv.org/abs/1808.00023>
- Equalized Odds in Fair Machine Learning. Emergent Mind. (2026, February 21). <https://www.emergentmind.com/topics/equalized-odds>
- Fraenkel, A. (2020). Fairness & Algorithmic Decision making. 5. parity Measures - Fairness & Algorithmic Decision Making. <https://afraenkel.github.io/fairness-book/content/05-parity-measures.html>
- Kleinberg, J., Mullainathan, S., & Raghavan, M. (2016, November 17). Inherent trade-offs in the fair determination of Risk Scores. arXiv.org. <https://arxiv.org/abs/1609.05807>
- Mayer, M., & Wilber, J. (2023, April). Equality of Odds: A Visual Introduction to Measuring and Mitigating Bias in Machine Learning. MLU-Explain. <https://mlu-explain.github.io/equality-of-odds/>
- Nigam, S. P. (2025, January 6). Predictive parity. AI Alchemy Hub. <https://www.aialchemyhub.in/ai-fairness-in-action/predictive-parity.php>
- Raftopoulos, G., Davrazos, G., & Kotsiantis, S. (2025, May 1). Evaluating fairness strategies in educational data mining: A comparative study of bias mitigation techniques. MDPI. <https://www.mdpi.com/2079-9292/14/9/1856>
- Saravanakumar, K. K. (2021, January 29). The impossibility theorem of machine fairness – a causal perspective. arXiv.org. <https://arxiv.org/abs/2007.06024>